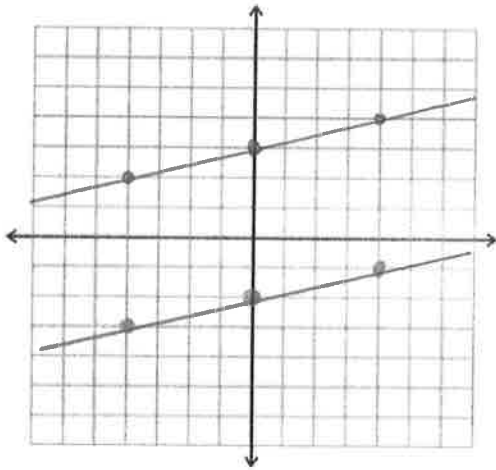


6.4 Parallel and Perpendicular Lines Worksheet

MPM1D

1. Graph each pair of lines on the same coordinate grid. Find their slopes and conclude whether the lines are parallel, perpendicular, or neither.

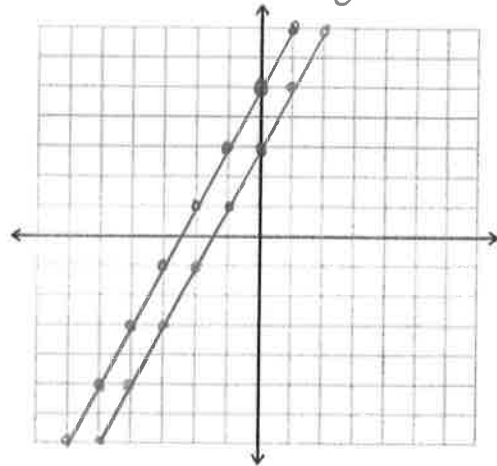
a) $y = \frac{1}{4}x - 2$ $y = \frac{1}{4}x + 3$



Parallel

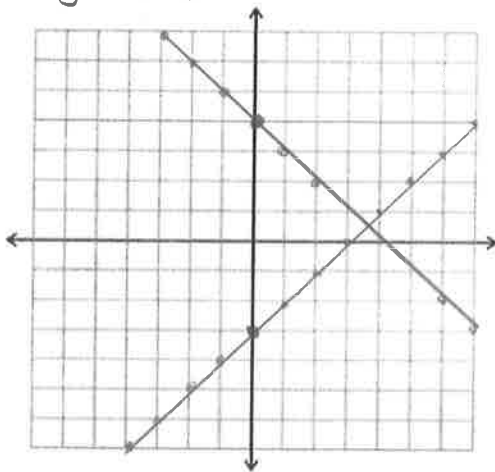
b) $y = 2x + 5$ $4x - 2y + 6 = 0$

$y = 2x + 3$



Parallel

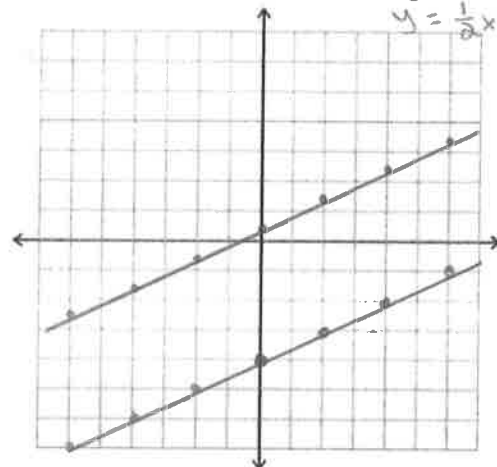
c) $x + y = 4$ $y = x - 3$
 $y = -x + 4$



Perpendicular

d) $y = \frac{1}{2}x - 4$ $x - 2y + 1 = 0$

$-2y = -x - 1$
 $y = \frac{1}{2}x + \frac{1}{2}$



Parallel

3. The slopes of two lines are given. Conclude whether the lines are parallel, perpendicular, or neither. Justify your answers

a) $m = \frac{2}{3}, m = \frac{4}{6} = \frac{2}{3}$

parallel

b) $m = \frac{3}{4}, m = -\frac{4}{3}$

perpendicular

c) $m = 2, m = -2$

neither

d) $m = 1, m = -1$

perpendicular

e) $m = \frac{1}{5}, m = 0.2$

parallel

f) $m = 2\frac{1}{4}, m = -\frac{4}{9}$
 $= \frac{9}{4}$

Perpendicular

4. What is the slope of a line that is parallel to each line?

a) $y = \frac{3}{5}x - 2$

$m = \frac{3}{5}$

b) $y = -x + 7$

$m = -1$

c) $2x - y + 3 = 0$

$y = 2x + 3$

$m = 2$

d) $4x + 3y = 12$

$y = -\frac{4}{3}x + 4$

$m = -\frac{4}{3}$

e) $y = 2$

$m = 0$

f) $x = -5$

undefined

5. For each line in question 4, give the slope of a perpendicular line.

a) $\perp m = -\frac{5}{3}$

b) $\perp m = 1$

c) $\perp m = -\frac{1}{2}$

d) $\perp m = \frac{3}{4}$

e) $\perp m = \text{undefined}$

f) $\perp m = 0$

6. Write the equations of two lines that are parallel to the line $3x - 6y - 5 = 0$

$-6y = -3x + 5$

$y = \frac{1}{2}x - \frac{5}{6}$

Parallel lines: ① $y = \frac{1}{2}x + 99$

② $y = \frac{1}{2}x + 87$

7. Write the equations of two lines that are perpendicular to the line $4x + y - 2 = 0$

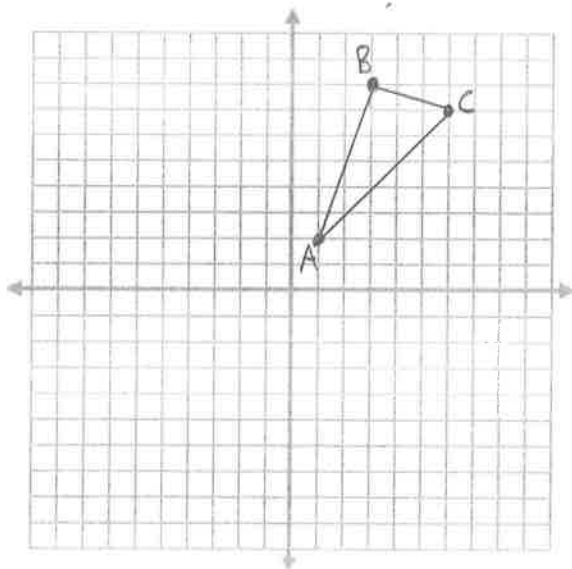
$$y = -4x + 2$$

Perpendicular Lines: ① $y = \frac{1}{4}x + 71$

② $y = \frac{1}{4}x + 87$

8. A triangle has vertices A(1, 2), B(3, 8), and C(6, 7).

a) Plot these points and draw the triangle.



b) Does this appear to be a right triangle? Explain.

It appears that at vertex B there might be a 90° angle.

c) Find the slopes of the line segments that form this triangle.

$$m_{AB} = \frac{8-2}{3-1}$$

$$= \frac{6}{2}$$

$$= 3$$

$$m_{BC} = \frac{7-8}{6-3}$$

$$= \frac{-1}{3}$$

$$m_{AC} = \frac{7-2}{6-1}$$

$$= \frac{5}{5}$$

$$= 1$$

d) Explain how the slopes can be used to conclude whether or not this is a right triangle. Is it?

The product of perpendicular slopes = -1.

$$m_{AB} \times m_{BC} = 3 \left(-\frac{1}{3} \right)$$

$$= -\frac{3}{3} = -1$$

~~-1~~ or AB and BC are perpendicular.

9. Determine whether or not the following sets of points form right triangles. Justify your answers with mathematical reasoning.

a) A(1, 1), B(-2, 5), C(3, -2)

$$m_{AB} = \frac{5-1}{-2-1}$$

$$= \frac{4}{-3}$$

$$= -\frac{4}{3}$$

$$m_{BC} = \frac{-2-5}{3-(-2)}$$

$$= -\frac{7}{5}$$

$$m_{AC} = \frac{-2-1}{3-1}$$

$$= -\frac{3}{2}$$

No slopes are negative reciprocals of each other, therefore the points do not form a right triangle.

b) P(2, 4), Q(-2, 2), R(5, -2)

$$m_{PQ} = \frac{2-4}{-2-2}$$

$$= \frac{-2}{-4}$$

$$= \frac{1}{2}$$

$$m_{QR} = \frac{-2-2}{5-(-2)}$$

$$= -\frac{4}{7}$$

$$m_{PR} = \frac{-2-4}{5-2}$$

$$= -\frac{6}{3}$$

$$= -2$$

$$m_{PQ} \times m_{PR} = \left(\frac{1}{2}\right)(-2)$$

$$= -\frac{2}{2}$$

$$= -1$$

$\therefore$ PQ and PR form a right angle.